



FACULTY OF COMPUTING

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SECJ3744 ENTERPRISE SYSTEMS DESIGN & MODELLING
SECTION 02

SAP and PPG Activities

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Introduction

This activity helps us understand the basic structure and main functions of the SAP Enterprise Resource Planning system. By exploring PPG Industries, we learn how digital systems like SAP support integration process and how global companies use this technology to manage operations and supply chains.

Key Observation from SAP

SAP is a comprehensive system that connects all areas of enterprise, such as finance, production, material management, sales, marketing, and human resources. It can help companies and organizations of all sizes and industries run their business profitably, adapt continuously, and grow sustainably. The core module of SAP is material management (MM), sales distribution (SD), finance (FI), production planning (PP), and human capital management (HCM).

SAP Process Flow

1. Procure-to-pay
 - This process starts with creating a purchase request → purchase order → goods receipt → invoice verification → payment
 - It ensures that purchasing and finance departments are integrated, reducing delays and improving financial accuracy.
2. Order-to-cash
 - Begin with a customer sales order → delivery → goods issue → billing → payment receipt
 - It improves customer satisfaction and cash flow visibility while connecting sales, inventory, and accounting modules.

Key Observation from PPG

How PPG integrates technology in its global operations?

PPG uses digital procurement platforms, data analytics, and ERP systems to manage suppliers, materials, and operations efficiently across its global network. The company has departments such as Global Category Management, Functional Support and Digital Procurement who work together to streamline processes, manage data, and create efficiencies throughout the system.

How do supply chains, production, or procurement rely on ERP-like systems?

ERP systems link the purchasing, inventory, and production line so that information flows smoothly between departments. This helps companies track materials, orders, and stock levels more efficiently. As a result, managers can make faster decisions, reduce delays, and improve overall efficiency.

Reflection

A company like PPG that uses SAP to connect all its core business processes in one system can improve transparency, coordination and data accuracy across its global operations. By using SAP, PPG can make better decisions, reduce manual work, and lastly achieve greater efficiency.

Process Flow example

The procure-to-pay process in SAP can be applied directly to PPG’s operations. For instance, when PPG sources raw materials such as pigment or resins, SAP’s integrated workflow can automate purchase orders, track inventory levels, verify supplier invoices, and manage payments. This integration would improve efficiency, reduce manual errors, and provide real-time data for procurement and finance departments.

SAP Enterprise System vs PPG Digital Business Model.

Aspect	SAP Enterprise System	PPG Digital Business Model
Purpose	Integrates all core business processes into one centralized platform	Use digital tools and analytics to manage global operations efficiently
Function	Stores and shares real-time data across departments for accuracy and consistency	Collects and analyzes data from suppliers and operations to support decision-making
Integration	Connects internal operations and external partners through ERP modules	Links suppliers, production, and sustainability efforts through digital procurement platforms

Conclusion

In conclusion, SAP and PPG demonstrate how digital systems enhance efficiency and data integration in global enterprises. By applying SAP processes, companies like PPG can achieve better transparency, coordination, and sustainability.

References

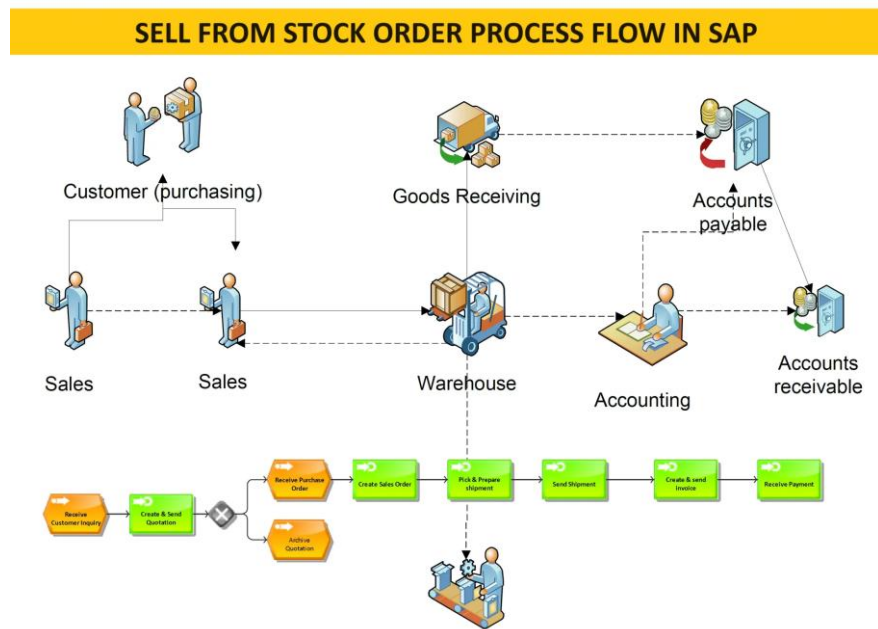


Figure 1: Example of SAP's process flow

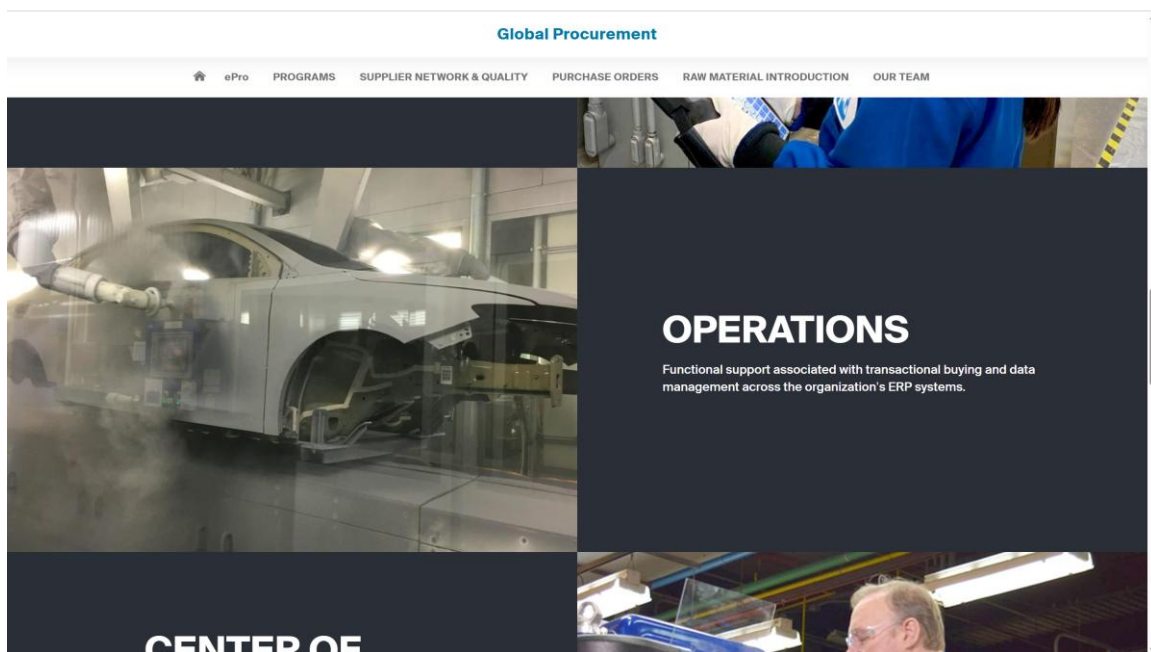


Figure 2: PPG uses ERP system in functional support associated with transactional buying and data management.

Evaluating supplier practices



Our procurement team continues to improve its approach to identifying improvement opportunities in our supply chain. We take a risk-based approach to supply chain management and focus on the industries and geographies most likely to have issues. PPG uses an external supplier screening tool to help us understand where sustainability risks are concentrated along our

supply chain. In 2024, we established a performance expectation for our supply base, requiring suppliers to achieve a minimum performance score across sustainability categories.

To drive continuous improvement, our procurement team works with suppliers to implement corrective action plans based on any deficiencies identified through our digital screening tool or supplier audits. Once corrective action plans are established, category managers are responsible for monitoring implementation and tracking performance improvements. This systematic approach continues to drive strong results, as evidenced by a significant year-over-year improvement in PPG's overall score from our external supplier screening tool. This in turn creates a competitive advantage for PPG through lower carbon footprint product formulations compared to our peers.



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Figure 3: PPG's procurement team chooses a risk-based approach in supply chain management